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Detection of Distributed Reflection Denial-of-Service (DRDoS) in Fog computing using Single Partitioned Q-learning with Explainability

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Abstract – Fog computing has emerged as an best solution to support real-time IoT applications by enabling data processing closer to end devices instead of depending entirely on cloud infrastructure. However, the distributed and resource-constrained nature of fog nodes makes them vulnerable to network attacks, particularly Distributed Reflection Denial-of-Service (DRDoS) attacks that can rapidly overload system resources and disrupt services. Although several existing detection approaches have been proposed, many of them either rely on predefined rules or lack transparency in explaining their decisions. To address these limitations, this paper proposes an explainable reinforcement learning based detection framework using Single Partitioned Q-learning (SPQL-X) for identifying DRDoS attacks in fog environments. The proposed approach learns traffic behavior dynamically and provides interpretable insights into attack detection decisions while maintaining efficient resource usage. Experimental results demonstrate that the model achieves improved performance in terms of detection accuracy, detection latency, energy efficiency and packet delivery ratio. The findings indicate that the proposed framework enhances attack detection capability while supporting stable and efficient fog-based network operations.

Index Terms – Man-in-the-Middle, Fog computing, Soft actor critic, Q-learning, Explainer.

I. INTRODUCTION

The IoT has grown rapidly because countless modern devices contain sensors to communicate automatically that supports real time monitoring, automation and reduced human errors [1]. By 2030, the number of IoT devices is expected to reach 34.6 billion and their data output to 246 zettabytes [2]. Fog computing helps to overcome this problem by processing data locally at fog nodes located near IoT devices and sends only the necessary or filtered data to the cloud, this reduces network latency, minimized

bandwidth consumption, enhanced scalability, improved energy efficiency, and support for real-time decision-making without waiting for cloud processing in applications such as smart cities, healthcare monitoring system and edge-based video analytics [3]. However, fog nodes operate in user-side environments that make them easy targets for attackers. Since they have limited processors and short battery life, they become vulnerable when too many requests arrive simultaneously, creating bottlenecks and resource starvation. They also face weaker security compared to cloud servers. Therefore, they encounter threats like Distributed Reflection Denial-of-Service (DRDoS), Distributed Denial-of-Service (DDoS), fake IP addresses, and botnet activity. These attacks cause sudden increases in network traffic, unusual patterns, unauthorized data access, throughput degradation, and service unavailability even before reaching the cloud. Early detection at this level is critical to quickly filter harmful traffic and stop attacks before they escalate. This requires smarter and efficient approaches with minimal resource usage.

Among the different cyber threats targeting fog computing environments, DRDoS attacks needs a special attention due to their unique amplification capabilities and stealth characteristics that makes traditional defenses ineffective. DRDoS is a type of DDoS attack, where the attacker tricks other server (reflectors) for sending a huge amount of traffic to the actual user by pretending to be their device. This causes the user to get overloaded with unwanted server responses [10]. In this attack, the attacker sends fake requests to many servers, and these servers unknowingly send much bigger responses to victim, that results in huge traffic load on victim devices [11]. Fog nodes serve as perfect reflectors because they depend on User Datagram Protocol (UDP) services for faster communication. UDP does not check sender authentication or request validation, which makes it easier for attackers to misuse it for reflection attacks. The attackers use them to reflect large response toward victims [12]. Unlike traditional DDoS attacks, DRDoS hides the attacker identity through reflectors,

creating detection challenges that is harder for fog devices to detect and handle these attacks [13].

Existing detection methods for DRDoS attacks include signature-based, anomaly-based and ML-based detection all fail fog computing requirements. Signature based detection looks for known attack patterns, but fails against new traffic patterns. Anomaly-based detection compares current traffic with normal behavior, but it takes more time and need large resources. ML-based detection can detect unknowns but hard to explain the reason [14]. These existing methods either needs more processing time or lack explainability that is not suitable for fog computing [15]. This gap motivates need for Single Partitioned Q-learning solution that provides efficient, accurate, and explainable detection for fog nodes.

Q-learning is a reinforcement learning method that learns through interaction with environment and receives reward or penalties [16]. It stores the actions in a Q-table with each state-action pair $Q(s,a)$ that lead to best move. Since fog devices needs continuous observation of traffic patterns and feedback receiving, Q-learning will help through simple numerical updates and doesn't require more computation to detect DRDoS attack patterns quickly [17]. Multi-partition methods split state spaces, increasing convergence time and memory overhead unsuitable for edge deployment. For DRDoS variants like DNS amplification, NTP reflection, and SSDP floods, single partition captures amplification factors, spoofed UDP patterns, and reflector reply patterns through single state learning. This enables fast detection of traffic floods (DNS/NTP) and protocol exploits (SSDP) with clear explanations through Q-value paths clearly – showing why traffic indicates DRDoS.

The remaining part of the paper is organized as follows: section 2 provides related work, section 3 explains about the proposed work, section 4 does mathematical modelling, section 5 presents the results obtained, and finally section 6 ends with the conclusion.

II. RELATED WORK

Khurshid et al. proposed a multi-layer automated DDoS mitigation framework spanning device, fog and cloud layers, using firewall rules at edge devices, statistical/behavioral analysis at fog nodes and deep packet inspection at cloud for stopping threats. Their system has three simple steps: devices block bad traffic using basic firewalls, fog computers check traffic more carefully to find attacks, and cloud gathers info to confirm and stop attacks. They also made a list of which defense steps can be automated like controls request speed, Synchronize sequence numbers (SYN cookies) validate the connection requests without storing state and traffic shaping. Their system works well to stop common attacks like Transmission Control Protocol Synchronize

(TCP SYN flood) that overloads connection handshakes, UDP flood and Internet Control Message Protocol results in ping requests (ICMP flood) but doesn't use any smart learning method with explainability to better detect and stop reflected DRDoS attacks. The systems dependence on predefined rules and multi-stage processing introduce delay unsuitable for fog computing real-time requirements, highlighting need for intelligent single-stage detection that rapidly identifies DRDoS amplification patterns at fog edge without delays.

Kang et al. focused on defending fog environments from DRDoS attacks, especially when the fog servers are low-power and cannot run heavy security systems. They studied how DRDoS over UDP quickly uses up fog server resources and bandwidth, and then proposed a mitigation model that uses a modified Snort UDP module with only a few selected functions enabled. Their system monitors UDP traffic on fog servers, uses simple pattern matching on fields like source port, UDP length and data content, and then sends suspicious packets across multiple fog servers using Anycast to share the filtering load. This approach has a simple structure and was able to keep more than 80% of normal packets delivered even during DRDoS, while using less CPU compared to a full Snort-based IDS/IPS. However, it still depends on fixed patterns and set UDP patterns, so it struggles when attackers change amplification vectors, packet sizes or data formats. It also does not learn from new DRDoS behaviors or give any explanation for why a packet is treated as attack traffic. This shows a need for intelligent, learning-based detection at fog level that can adapt to evolving DRDoS traffic and provide clear reasons for each detection decision, which is what Single Partitioned Q-learning aims to address.

Diogo Vinicius Martins da Cruz et al. proposed Fuzzy Inference System (FIS) for identifying DDoS attacks in fog computing by analyzing power usage patterns of Message Queuing Telemetry Transport (MQTT) brokers. Since fog devices with limited resources near edge users, frequent malicious traffic like SYN flood, ICMP flood and Large Packet Flood significantly increase their power usage. The authors used fuzzy logic to classify system states based on parameters such as CPU load, memory usage, network traffic and energy consumption levels with simple words like Low, High, Very High Consumption and Very Short Time (VST), Long Time (LT) for time, creating 25 fuzzy rules to classify attacks and give strength level. Their results showed that SYN flood caused highest jump (31% increase to 2.45 Wh) making it good indicator for DDoS in fog platforms. However, this approach depends on predefined fuzzy rules and fixed thresholds like consumption [0-3 Wh] that may become outdated and limits its ability to adapt to dynamic and evolving attack patterns and misses basic network spoofing detection and cannot self-learn new attack

behaviors, highlighting the need for adaptive and self-learning approaches.

Muhammad Rashid Minhas et al. proposed a fog-based architecture called Fog-based Online Self-adaptive Framework for Attack detection (F-OSFA) for detecting both traditional and zero-day DDoS attacks in fog environments [23]. The system uses a combination of Convolutional Neural Network (CNN) and Decision Tree for previously existing attacks, a Contractive Autoencoder (CAE) for zero-day attack (new attacks never seen before) detection, and a signature-based analyzer to identify DDoS traffic that looks like normal traffic. The model was tested and achieved high accuracy of 99.65% on CICDDoS2019 showing fog-level intelligence works well for DDoS. However, the method needs a large training data just to detect anomalies only, but doesn't take any decision action after finding attacks. Also high dimensional datasets (80+ features) makes it heavy for fog nodes with limited. This shows the need for solution which can make decisions and take actions after the anomaly detection.

Gudla et al. designed a Deep Intelligent DDoS Attack Detection Scheme (DI-ADS) for fog-based IoT applications using a Deep Neural Multilayer Perceptron (DNMLP) model at the fog layer to classify IoT device behavior as normal or attacker based on features like packet flow, protocol type and traffic statistics. The framework compares several models (DNMLP, LSTM, SVM, KNN, LR, RF) on the DDoS-SDN dataset containing TCP SYN, UDP flood and ICMP attacks, and selects DNMLP, which achieved 99.44% accuracy and lower behavior detection time than LSTM, especially as the number of fog nodes increases, showing that deep models at fog level can give both fast response and high accuracy compared to traditional ML. However, DI-ADS performs supervised binary classification, so it depends heavily on large labelled datasets and cannot easily handle new or evolving DRDoS patterns without retraining, and the DNN acts as a black box without giving reasons for why a device is marked as attacker, which increases computational cost and reduces transparency at resource-limited fog nodes; this indicates the need for approaches that work with less labelled data, learn continuously during operation, and provide explainable decisions.

III. PROPOSED INTERPRETABLE SOFT ACTOR CRITIC Q-LEARNING FRAMEWORK

The proposed Single Partition Q-learning framework works in the fog layer to detect DRDoS attacks with clear explanations. It maps all traffic features into one state space using a single Q-table, helping learn attack patterns under uncertainty. The agent observes states, selects mitigation actions, and provides explainable reasons.

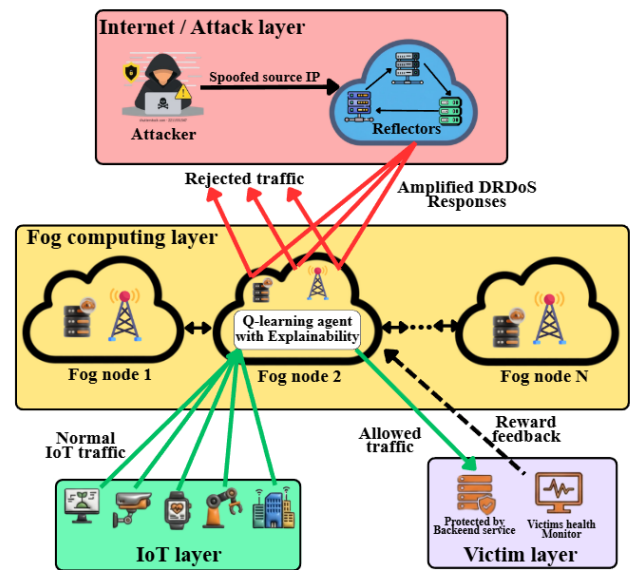


Figure 1: Proposed Single Partition Q-learning architecture for explainable DRDoS detection in fog computing.

Figure 1 shows the proposed DRDoS detection system organised in four main layers: the IoT layer, the fog computing layer, the Internet layer and victim layer.

At the IoT layer, many devices such as sensors, cameras and controllers send their regular data (measurements, video, control signals) towards nearby fog nodes using lightweight protocols over Wi-Fi, 5G or wired links. This traffic is mostly normal and represents the genuine service requests and responses of IoT applications. At the same time, in the Internet/attack layer, an attacker sends small spoofed UDP requests to reflector servers (for example DNS, NTP or SSDP), which then generate much larger responses towards the victim's IP address. These amplified responses also pass through the same fog nodes, so each fog node receives a mixture of normal IoT traffic and potential DRDoS amplification traffic on its input links.

In the fog layer, the proposed Single Partition Q-learning based node observes this mixed traffic. A monitoring module summarises the behaviour of flows into simple features and the Q-learning agent treats each feature vector as a state. For each state, the agent uses a single Q-table to choose an action for specific flows or ports. The mitigation part of the fog node then enforces these decisions so that suspected DRDoS amplification traffic is dropped, while the normal IoT traffic is preserved.

Finally, only the safe, filtered traffic reaches the victim layer. The fog node gets feedback from how well the victim is working (like its speed, packet loss, or availability). This feedback acts as a reward score that updates the Q-learning table, helping the fog node get smarter over time, it learns exactly which actions block DRDoS attacks best while letting normal IoT traffic through smoothly.

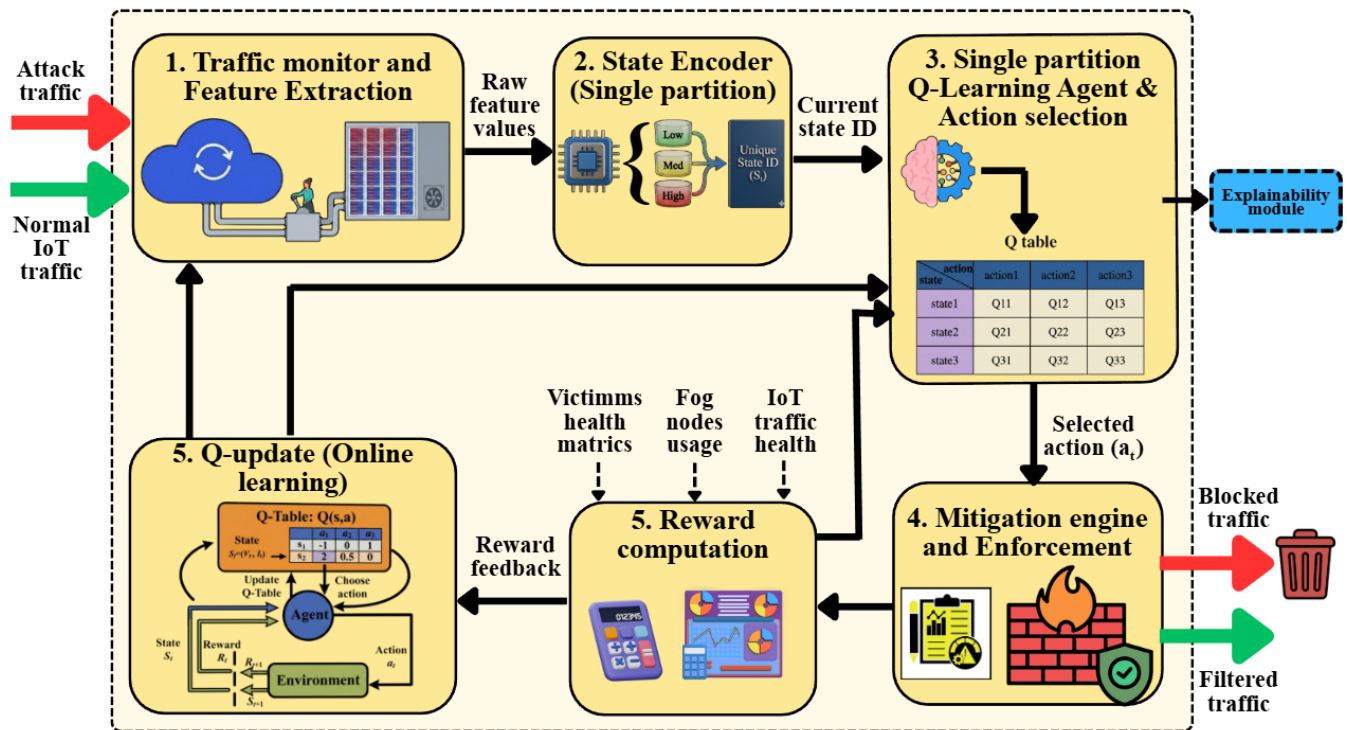


Figure 2: Proposed fog node architecture for Single Partition Q-learning based DRDoS detection, mitigation, and explainability.

Figure 2 shows the detailed internal architecture of a single fog node, where the core DRDoS detection and mitigation logic is executed. The fog node receives mixed traffic from IoT devices and from Internet reflectors and processes it through five interconnected modules in a continuous loop.

Incoming packets and flows are first captured by the Traffic Monitor & Feature Extractor module, which groups traffic into short time windows (1-2 seconds). It computes key DRDoS indicators such as UDP packet/byte counts per port group (DNS port 53, NTP port 123, SSDP port 1900), response-to-request byte ratios for amplification detection, source/victim IP diversity, basic spoofing signs like TTL anomalies, and local fog resource usage (CPU, memory, outbound bandwidth). This produces a compact feature vector every window that captures whether the current traffic looks normal or suspicious.

The feature vector feeds into the State Encoder (Single Partition), which discretises each feature into bins (low/medium/high) and maps them to a single global state ID, ensuring all DRDoS types and normal patterns share one unified state space without separate partitions. The Single Partition Q-Learning Agent & Action Selection then uses this state ID to look up the single Q-table (rows: states, columns: actions like allow/rate-limit/drop). It selects the best action via ϵ -greedy policy (mostly optimal, sometimes explore) based on learned Q-values.

The selected action goes to the Mitigation Engine & Enforcement, which applies it instantly with low-overhead rules: updating ip tables/nf tables firewalls, setting per-port/flow rate limits, dropping matching traffic, or tagging for logs. Normal IoT traffic gets forwarded, while DRDoS flows are rejected. Simultaneously, the Explainability Module reads the state, action, Q-values, and recent rewards to generate simple human-readable notes, such as "High DNS amplification and rising victim loss $\rightarrow$ drop DNS flows; this action historically improves rewards fastest."

After enforcement, the Reward Computation module evaluates outcomes from three sources: victim health (low utilisation, good throughput), IoT traffic success rates, and fog resource loads. These combine into a scalar reward that updates the Q-table entry for that state-action pair via the standard Q-learning update. Over repeated episodes from normal operation, attack onset, to stable mitigation, the agent learns optimal policies for new DRDoS variants without manual signatures, adapting continuously while providing transparent explanations.

Algorithm 1 describes the working of the proposed Single Partitioned explainable Q-learning framework for detecting DRDoS attacks in fog computing. It explains how traffic features are analyzed, states are formed, and mitigation actions are selected using reinforcement learning. The algorithm also provides clear reasoning for each detection decision.

Algorithm 1: Working of the proposed Explainable Q-learning framework for DRDoS detection in Fog Computing

1. **Start**
2. **Input:** Network traffic stream T , Feature extraction function F , State encoding function φ , Action space A , Learning rate α , Discount factor γ , Initial exploration rate ε_0 , Minimum exploration rate ε_m , Decay factor δ , Fog threshold θ_f , Victim threshold θ_v , Maximum iteration I_{max} .
Output: Mitigated traffic stream T_{safe} , Optimal policy $\pi^*(s)$, Explainability logs X_{local} , X_{global} , Performance metrics R , Explainability artifact set E .
3. **Define** state space $S = \emptyset(R^4)$,
 Action space $A = \{a_1, a_2, a_3, a_4\}$ (where a_1, a_2, a_3, a_4 are Allow traffic, Throttle rate, drop packets, Degrade service)
4. **Initialize** Q-function $Q(s, a) \leftarrow 0 \forall (s, a) \in S \times A$,
 Explainability set $X \leftarrow \emptyset$ ($X \subseteq S \times A \times R \times R^3$ where R and R^3 are reward and system metrics respectively),
 $\varepsilon \leftarrow \varepsilon_0$ ($\varepsilon_{min} \leq \varepsilon \leq 1$), $\delta \in (0, 1)$
 iteration index $t \leftarrow 0$.
5. **Set** $\alpha \in (0, 1)$, $\gamma \in (0, 1)$, $\delta \in (0, 1)$ and observe initial traffic $T_0 \in R^n$ that contains packet rates, sizes and timing.
6. **Compute** feature vector:
 $x_0 = F(T_0) = [r_0, d_0, b_0, c_0]^T \in R^4$,
 where r_0, d_0, b_0, c_0 are request rate, delay, bandwidth, CPU usage. This converts raw traffic into structured data.
7. Map feature space to state space: $s_0 = \varphi : R^4 \rightarrow S$
8. **while** $t < I_{max}$ do:
9. $x_t = F(T_t) = [r_t, d_t, b_t, c_t]^T$ //extract feature vector
10. $\hat{x}_t = (x_t - \mu) \div \sigma$ //normalize features with mean and standard deviation
11. $s_t = \varphi(\hat{x}_t) \in S$ //discretize and state encoding
12. $r \sim U(0, 1)$ // sample random variable
13. **If** ($r \leq \varepsilon$) then
 $a_t \leftarrow a, a \in A, P(a) = \frac{1}{|A|}$ //random action selected
14. **else**
 $a_t \leftarrow \arg \max Q(s_t, a)$ //selects best action
15. **End if**
16. Apply mitigation operator: $M_t = g(a_t, T_t)$ //where g is a mitigation function that applies action to traffic.
17. Measure:
 $C_t = cpu(M_t) \in R^+$
 $B_t = bw(M_t) \in R^+$
 $H_t = succ(M_t) \in [0, 1]$
 $L_t = loss(M_t) \in [0, 1]$
19. Reward formulation:

$$R_{system} = \begin{cases} H_t & \text{if } C_t \leq \theta_f \wedge B_t \leq \theta_v \\ -(C_t + B_t + L_t) & \text{otherwise} \end{cases}$$
20. Explainability regularization:
 $R_{explain} = -\lambda \cdot Entropy(\pi(s_t))$
 where $Entropy(\pi(s_t)) = -\sum_a P(a|s_t) \log P(a|s_t)$
21. Final reward: $r_t = R_{system} + R_{explain}$
22. Observe next traffic window: T_{t+1}
23. Extract next feature vector: $x_{t+1} = F(T_{t+1})$
24. Normalize and encode next state:
 $s_{t+1} = \varphi((x_{t+1} - \mu) \div \sigma)$
25. Compute optimal future return:
 $Q_{max} = \max Q(s_{t+1}, a)$
26. Update Q-function (Bellman optimally):
 $Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[r_t + \gamma \times Q_{max} - Q(s_t, a_t)]$
27. Feature attribution (Local explainability):
28. For each feature x_i :
 $I_i = |Q(s_t, a_t) - Q(s_t^{-i}, a_t)|$
 where s_t^{-i} = state with feature i masked
29. Select top k influential features: $F_{top} = TopK(I_i)$
30. Confidence Score:
 $Conf_t = \frac{Q(s_t, a_t)}{\sum_{a \in A} Q(s_t, a)}$
31. Counterfactual Explanation:
32. For each $x_i \in F_{top}$:
 $x'_i = x_i - \Delta$
 $s'_i = \varphi(x'_i)$
 $a'_i = \arg \max Q(s'_i, a)$
33. If $a'_i \neq a_t$: Store counterfactual:
 "Had x_i been x'_i , action would be a'_i "
34. Append local explanation:
 $X_{local} \leftarrow X_{local} \cup \{ \{s_t, a_t, r_t, F_{top}, Conf_t, C_t, B_t, L_t\} \}$
35. Global explainability update
 Update frequency map: $Freq(s, a) = Freq(s, a) + 1$
 Update global importance: $GI_i = \sum_t I_i$
36. Store dominant policy rules in X_{global}
37. $s_t \rightarrow s_{t+1}$ //state transition
38. $\varepsilon \leftarrow \max(\varepsilon_m, \varepsilon \times \delta)$ //update exploration rate
39. $t \leftarrow t + 1$ //Increment iteration counter
40. **End while**
41. Derive optimal policy:
 $\pi^*(s) = \arg \max Q(s, a)$
42. **For** each interface window: $\forall T_i \in T$ do,
43. $x_i = F(T_i)$ //extract feature vector
44. $s_i = \varphi(x_i)$ //encode system state
45. $a_i = \pi^*(s_i)$ //select optimal action
46. $M_i = g(a_i, T_i)$ //applying mitigation mapping
47. **End for**
48. Aggregate mitigated traffic: $T_{safe} = \cup_i M_i$
49. **Compute:**
 $Acc = \frac{TP}{TP+FN}$ //detection accuracy
 $FPR = \frac{FP}{FP+TN}$ //false positive rate
50. Compute mitigation latency: $L = t_{mit} - t_{det}$
51. **If** ($Acc < \tau_{acc} \vee FPR > \tau_{fpr}$) then
 Reinitialize $\{\alpha, \gamma, \varepsilon\}$
 Relearn $Q: S \times A \rightarrow R$
52. **End if**
53. Store learned Q-function: $Q \rightarrow M_{persistence}$
54. Generate performance metric set and Explainability:
 $R = \{Acc, FPR, L\}$
 $E = \{X_{local}, Conf, X_{global}\}$
55. **Output:** $\langle T_{safe}, \pi^*, X_{local}, X_{global}, R, E \rangle$
56. **Stop**

IV. RESULTS

The proposed Single Partitioned Q-Learning with Explainability (SPQL-X) framework is compared with two existing DRDoS detection approaches, namely the multi-layer fog mitigation framework (MLFM) by Khurshid et al. and the modified Snort-based UDP reflection detection model (MSUR) by Kang et al. The evaluation considers performance parameters such as detection accuracy, detection latency, energy consumption and packet delivery ratio (PDR). The simulation setup is defined as follows: number of fog clusters = 2, number of fog nodes varying from 50 to 500, processing capacity = 1000–3000 MIPS, RAM = 8 GB, storage = 1 TB, bandwidth = 10 Gbps, simulation time = 3000 seconds, traffic load ranging from 500 to 8000 packets per second, amplification attack strength ranging from 20% to 100%, packet size = 300 KB, initial energy of each node = 100 J, transmission energy = 0.65 W, reception energy = 0.39 W, and maximum learning iterations = 10,000. DRDoS traffic is generated using reflected UDP amplification through distributed reflector nodes, and the behavior of the detection framework is observed under increasing traffic load, amplification intensity, infrastructure size, and attack duration.

Model scalability under high traffic

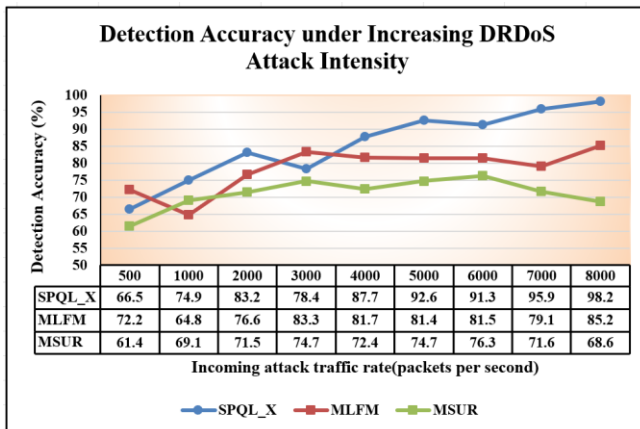


Figure 2: Detection Accuracy vs Traffic Load

Figure 2 presents Detection Accuracy with Traffic Load, where the X-axis represents DRDoS traffic rate (ps) and the Y-axis represents detection accuracy (%). The experiment was conducted by gradually increasing reflected UDP traffic and allowing the learning model to observe evolving patterns across fog nodes. The SPQL-X framework adapts to traffic behavior through reinforcement learning updates, which enables reliable identification of amplification signatures during heavy traffic conditions. The MLFM approach suffers accuracy reduction due to dependence on predefined mitigation stages that limit adaptability to new attack signatures, while the MSUR model shows misclassification because it relies on static UDP reflection thresholds.

Detection capability

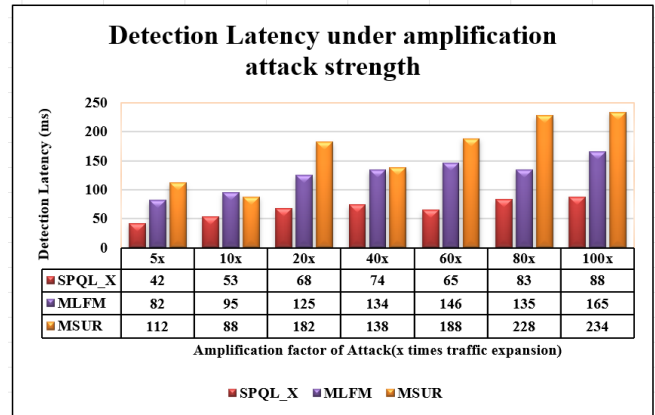


Figure 3: Detection latency vs Amplification factor

Figure 3 shows Detection Latency under Amplification Attack Strength, where the X-axis indicates amplification factor (%) and the Y-axis represents detection delay (ms). The experiment introduces increasing reflection intensity from distributed reflectors and records the time required to detect abnormal traffic flows. The SPQL-X model identifies attack behaviour early because the Q-learning policy continuously evaluates state transitions and learns amplification characteristics during operation. The MLFM framework records higher delay because communication overhead between defense modules slows decision propagation, while MSUR detection time increases due to packet-level rule verification across multiple flows.

Energy efficiency

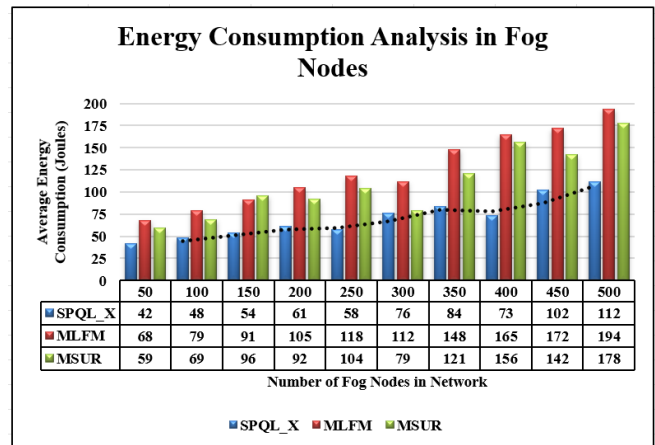


Figure 4: Energy consumption vs Fog nodes

Figure 4 illustrates Energy Consumption with Increasing Fog Nodes, where the X-axis represents number of fog nodes (50–500) and the Y-axis represents average energy consumption (J). The evaluation distributes DRDoS traffic across a growing fog infrastructure and measures the processing energy required for monitoring and mitigation. The SPQL-X framework operates with lightweight Q-table updates and reduced computational overhead, which helps preserve

energy while maintaining detection operations in distributed nodes. The MLFM framework consumes more energy as coordination between distributed mitigation layers increases processing workload, while the MSUR model performs continuous traffic scanning that raises processing cycles and power usage.

Network Stability during Prolonged DRDoS Attacks

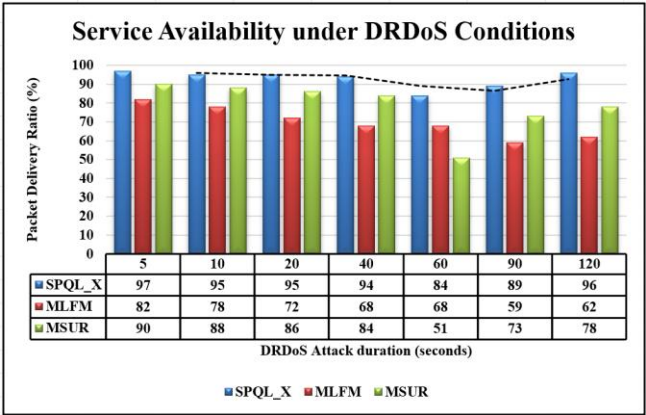


Figure 5: CPU Utilization vs Traffic Load

Figure 5 presents Packet Delivery Ratio during DRDoS Attack Duration, where the X-axis represents attack duration (seconds) and the Y-axis represents packet delivery ratio (%). The simulation introduces DRDoS traffic for longer durations while observing the number of legitimate packets successfully delivered through the fog network. The SPQL-X framework maintains stable packet delivery since the learning agent filters abnormal traffic flows early and protects normal communication paths. The MLFM framework shows reduced delivery performance since mitigation triggers activate only after traffic crosses certain thresholds, while the MSUR model experiences service degradation when rule processing delays allow attack traffic to occupy bandwidth.

V. CONCLUSION

This paper addresses a significant security challenge in fog computing environments, namely the detection of Distributed Reflection Denial-of-Service (DRDoS) attacks that can severely affect network performance and service availability. To address this problem, an explainable reinforcement learning-based framework using Single Partitioned Q-learning (SPQL-X) is proposed for efficient detection and mitigation of DRDoS attacks in fog architecture. The experimental results demonstrate that the proposed model performs effectively across important performance metrics such as detection accuracy, detection latency, energy consumption and packet delivery ratio under varying attack intensities and network conditions. The results also indicate that the SPQL-X framework provides better adaptability and scalability compared to traditional rule-based (MSUR) and signature-based (MLFM) detection

approaches. In addition, the explainability feature improves transparency by providing insights into the detection decisions made by the model. Overall, the proposed approach enhances network stability and supports reliable fog-based security systems, and it can be extended in future work to detect other potential attacks such as advanced DDoS variants, replay attacks, and coordinated network threats.

REFERENCES

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